

مجالات التخصص لاختبارات المعيارية للعام 1447 هـ

GKU	Topics and explanation		Subject	Inst. Theoretical	Practical
1	Basic Analysis and Algorithmic Methods	Fundamental concepts in analyzing algorithms, including time and space complexity, Big O notation, and optimization strategies. Example: Analyzing the time complexity of a sorting algorithm like QuickSort	الخوارزميات	فاروق عبدالله	سيد رشاد
				مروى بلقاسم	مروى بلقاسم/ اسراء السياط
2	Algorithms, Design, and Development	Theoretical and practical aspects of algorithm design, including searching, sorting, and optimization techniques. Example: Designing a binary search algorithm to efficiently find an element in a sorted array	الخوارزميات	فاروق عبدالله	سيد رشاد
				مروى بلقاسم	مروى بلقاسم/ اسراء السياط
3	Program Representation, Language, and Interpretation	How programming languages represent data and control structures, including syntax, semantics, and execution models. Example: Understanding how a high-level language like Python is translated into machine code through an interpreter	مقدمة في البرمجة	لافي العنزي	سيد رشاد
				رندا اللافي	شمس السلامي
4	Fundamental Programming Concepts	Basic programming constructs such as variables, control structures, functions, and error handling. Example: Writing a simple program that uses loops and conditionals to find prime numbers	مقدمة في البرمجة	سامي العنزي	سيد رشاد
				عائدة الصغير	ايمان حموده
5	Basic Type Systems	Understanding data types, type safety, type inference, and polymorphism in programming languages. Example: Demonstrating static vs. dynamic typing with examples in languages like C (static) and Python (dynamic)	مقدمة في البرمجة	نافع العنزي	سيد رشاد
				عائدة الصغير	ايمان حموده
6	Object-Oriented Programming	Principles of OOP such as encapsulation, inheritance, and polymorphism, focusing on design patterns and software reusability. Example: Creating a class hierarchy for shapes (e.g., Circle, Rectangle) with methods for calculating area and perimeter	البرمجة الشيئية	عبدالقادر المرس	دانش منظور
				شمس السلامي	شمس السلامي

7	Fundamentals of Data Structures	Core data structures (arrays, lists, stacks, queues) and their implementation and application in algorithms. Example: Implementing a stack using an array and demonstrating push/pop operations	هياكل البيانات	اشرف ميلاد	دانش منظور
				ايمان حموده	ايمان حموده
8	Database Systems and Data Models	Principles of database design, normalization, querying languages (SQL), and data integrity. Example: Designing a relational database schema for a library, including tables for books, authors, and borrowers	مقدمة في قاعدة البيانات	السعيد مشاحيت	محمد رافي
				أسماء الهشمي	أسماء الهشمي
9	Software Design, Construction, and Verification	The methodologies for designing, building, and verifying software to ensure quality and maintainability. Example: Using UML diagrams to design a software application and writing unit tests to verify functionality	هندسة البرمجيات	ماجد نواز	ماجد نواز
				وعد الدندني	وعد الدندني/اسراء السياط
10	Software Processes, Software Management	Software development life cycles, project management, and methodologies like Agile and Waterfall. Example: Implementing an Agile process by creating user stories and iterations for a software project.	هندسة البرمجيات	شوقي عباد	ماجد نواز
				وعد الدندني	وعد الدندني/اسراء السياط
11	Concurrency, Scheduling, Dispatch	Techniques for managing concurrent processes, including thread management, scheduling algorithms, and synchronization issues. Example: Implementing a producer-consumer problem using threads and synchronization mechanisms like semaphores	أنظمة التشغيل	جمال خميس	أبو طه زماني
				سناء بخيت	سناء بخيت
12	Operating System Principles	Core concepts of operating systems including process management, memory management, file systems, and resource allocation. Example: Process scheduling using Round Robin or First-Come, First-Served algorithms	أنظمة التشغيل	جمال خميس	أبو طه زماني
				سناء بخيت	سناء بخيت
13	Reliable Data Delivery, Routing, and Networking	Principles of networking, including protocols, data transmission, routing algorithms, and ensuring reliable communication. Example: Understanding how TCP ensures reliable data delivery over IP networks	الشبكات	مشاري حويتم	محمد عياز خان
				أحلام الفطناسي	أحلام الفطناسي

14	Introduction to Networking and Communication	Basics of computer networks, including types of networks, protocols, and network security. Example: Setting up a simple client-server architecture	الشبكات	مشاري حويتم	محمد عياز خان
				أحلام الفطناسي	أحلام الفطناسي
15	Graphs, Trees, and Discrete Probability	Data structures (graphs and trees) and their applications, along with discrete probability theories used in algorithms. Example: graphs and trees and discrete probability	رياضيات متقطعة	سالم حمود / خالد الحربي	ايمان حموده/ راضية المبروك
16	Proof Techniques and Basics of Computation	Mathematical reasoning in computer science, proof techniques such as induction, and basic concepts of computability. Example: Proving that the sum of the first nn natural numbers is $n(n+1)/2$ using mathematical induction	رياضيات متقطعة	خالد الحربي / سالم حمود	ايمان حموده/ راضية المبروك
17	Basic Logic, Sets, Relations, and Functions	Mathematical foundations of computer science, including set theory, logic, functions, and their applications. Example: Using Venn diagrams to illustrate set operations (union, intersection) and their properties	رياضيات متقطعة	سالم حمود / السعيد مشاحيت	ايمان حموده/ راضية المبروك
18	Machine Organization on the Basis of Architecture	Understanding the structure and organization of computer hardware, including CPU architectures and instruction sets. Example: Describing the fetch-decode-execute cycle in a CPU and how it processes instructions	معماريات الحاسب	فراس علان	أبو طه زماني
				راضية المبروك	عائدة الصغير
19	Architecture of Memory Systems	Understanding memory hierarchies, cache design, virtual memory, and how they affect performance. Example: Explaining the differences between RAM and cache memory and their roles in a computer system	معماريات الحاسب	فراس علان	أبو طه زماني
				راضية المبروك	عائدة الصغير
20	Digital Logic, Digital Systems	Foundations of digital circuits, logic gates, and the design of digital systems and processors. Example: Designing a simple combinational circuit using AND, OR, and NOT gates to implement a specific logic function	تصميم المنطق والنظم الرقمية	ماجد نواز	سيد مطيع
				رندا اللافي	أحلام الفطناسي